

MUSEMIND

AI PERSONAL JOURNAL

Write it. Understand it. See the pattern.

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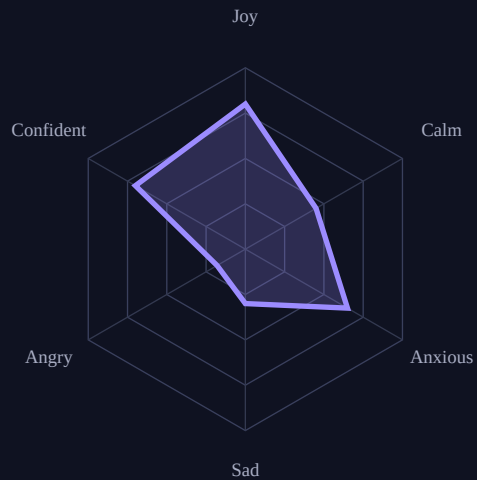
01 — INTRODUCTION

The idea

- Students and professionals often experience stress, anxiety and emotional pressure from academic, personal and social situations.
- Journaling gives users a private space to express thoughts and reflect.
- MuseMind adds an intelligence layer so a journal is not only stored, but analyzed for meaningful emotional signals.
- The current prototype combines a pretrained GoEmotions-based transformer, MuseMind mapping logic and visual emotional history.

From simple journal → emotion-aware experience

Illustrative emotion profile



02 — LITERATURE REVIEW

Reference	Observation
Reflectly	AI journaling and mood tracking, mainly focused on mobile use.
Zenie AI / Aksha AI	Emotion-focused journaling with less emphasis on transparent analytics.
Journaling research	Journaling combined with reflection and feedback can support emotional awareness.
Gap	Opportunity for a simple web journal combining emotion detection, history and visual insights.

03 — PROBLEM STATEMENT

THE OLD EXPERIENCE

- Journal stores the entry.
- One mood label can oversimplify mixed feelings.
- Little visibility into emotional change.

MUSEMIND RESPONSE

- Use a GoEmotions-based transformer.
- Preserve multiple emotion signals.
- Turn results into visual insights.

CURRENT SCOPE

- Functional browser prototype.
- Data currently uses localStorage.
- Backend is future work.

The goal is not just **“you are happy.”** The goal is to show **what emotional signals are present and how they relate to the journal.**

04 — PROPOSED SOLUTION



JOURNAL

- Natural text input
- Drafts and entries
- Word / character information

EMOTION INTELLIGENCE

- Primary + secondary
- 28 standard labels
- Mixed emotions
- Confidence scores

INSIGHTS

- Emotion radar
- Sentence analysis
- Emotion journey
- Mood score
- Supporting signals

05 — OBJECTIVES

01

Build a usable web journaling and reflection interface.

02

Integrate a legitimate pretrained GoEmotions-based emotion model.

03

Support multiple emotions and sentence-level analysis.

04

Visualize emotional history with charts, radar and comparisons.

05

Keep MuseMind categories while preserving standard NLP labels.

06

Keep the architecture modular for future backend and AI improvements.

06 — PROPOSED METHODOLOGY

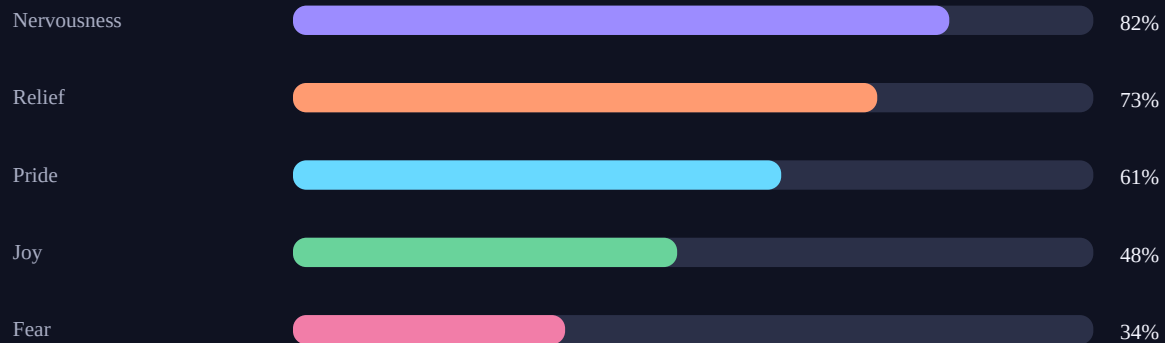
- Step 1 — User logs in or creates a profile.
- Step 2 — User writes a journal entry.
- Step 3 — Language routing selects transformer or fallback.
- Step 4 — RoBERTa returns multi-label GoEmotions probabilities.
- Step 5 — All 28 outputs are retained; meaningful results are selected for display.
- Step 6 — Standard labels are mapped to MuseMind categories.
- Step 7 — Sentence-level results can build an emotion journey.
- Step 8 — Results are stored with model/source information and visualized.



06A — THE NEW AI LAYER

DATASET	GoEmotions — Google Research; approximately 58K Reddit comments; 28 labels.
MODEL	MicahB/roberta-base-go_emotions — RoBERTa-based pretrained/fine-tuned model.
TASK	Multi-label emotion classification.
INFERENCE	Browser-based Transformers.js; loaded dynamically when analysis is requested.
FALLBACK	Existing MuseMind rule-based engine for Hindi/Hinglish or unavailable transformer analysis.

Illustrative model-output view



The chart above is an illustrative UI example, not a MuseMind accuracy benchmark.

06B — 28 LABELS → MUSEMIND

MuseMind	GoEmotions labels
Happy	joy, amusement, love, gratitude, admiration, approval, caring
Excited	excitement, surprise, desire
Sad	sadness, grief, disappointment, remorse
Angry	anger, annoyance, disgust, disapproval
Anxious	fear, nervousness, embarrassment, confusion
Calm	relief, neutral
Confident	pride, optimism, realization, curiosity

Example: “I was nervous before my presentation, but afterwards I felt proud and relieved.” → Nervousness → Anxious | Pride → Confident | Relief → Calm.

07 — CURRENT OUTCOMES

WORKING

- Login/profile flow
- Journal creation
- Dashboard + history
- Responsive interface

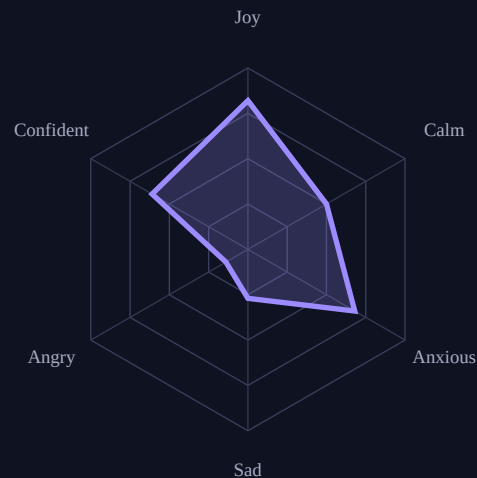
AI / NLP

- GoEmotions transformer
- 28-label storage
- Primary + secondary
- Mixed emotion
- Sentence analysis

INTELLIGENCE

- Emotion radar
- Emotion journey
- Mood score
- Baseline + comparison
- Supporting signals

Illustrative emotion profile



Illustrative mood trend



Illustrative UI example — not measured project data.

Illustrative charts show how the prototype presents emotion/mood information; they are not measured project statistics.

07A — FEATURE SHOWCASE

MIXED EMOTION

- Primary: Nervousness
- Secondary: Pride
- Additional: Relief
- Status: Mixed emotion

EMOTION JOURNEY

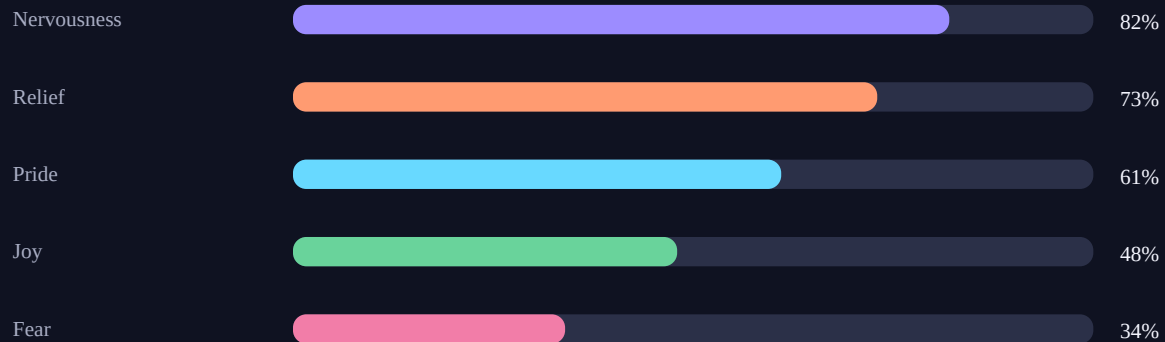
- Before → Nervousness
- During → uncertainty
- After → Pride
- Final → Relief

TEXT EVIDENCE

- Original journal is preserved.
- Supporting signals can be highlighted.
- Not presented as model reasoning.

A strong prototype should feel like an **experience**, not just an output box.

Illustrative model-output view



07B — WHAT IS ACTUALLY IMPLEMENTED?

Feature	Status
Login / profile	Implemented
Journal + local persistence	Implemented
GoEmotions RoBERTa integration	Implemented
28-label storage	Implemented
Multi-emotion detection	Implemented
Sentence-level analysis	Implemented
Emotion radar / insights	Implemented
Re-analysis of older entries	Implemented
MongoDB / Firebase backend	Future work
Production authentication	Future work
Dedicated multilingual transformer	Future work

08 — FUTURE WORK

01 — Backend

Move persistent data from localStorage to a real backend/database.

02 — Authentication

Add production-grade authentication and secure server-side handling.

03 — Multilingual NLP

Add a genuinely multilingual emotion model for Hindi/Hinglish.

04 — Evaluation

Create a labeled MuseMind test set and report real precision, recall and F1.

05 — Voice Journaling

Add speech-to-text for easier input.

06 — Personalization

Use longer-term history for stronger reflections and pattern detection.

09 — CONCLUSION

MuseMind has evolved from a basic rule-based journaling prototype into a more structured emotion-aware web application.

- A pretrained RoBERTa model fine-tuned on GoEmotions is integrated for English emotion classification.
- Multiple meaningful emotion signals can be preserved instead of forcing one mood label.
- The 28 standard labels are retained and mapped into user-friendly MuseMind categories.
- The analysis now includes confidence, mixed emotions, sentence-level results, emotion journey, supporting signals, mood scoring and history comparison.
- Current persistence is browser localStorage; production backend and stronger multilingual NLP remain future work.
- MuseMind is an informational prototype, not a clinical assessment.

From “write a journal” to “understand your emotional pattern.”

10 — REFERENCES

- Google Research — GoEmotions dataset and research paper.
- Hugging Face — MicahB/roberta-base-go_emotions model.
- Hugging Face — Transformers.js documentation.
- Chart.js — browser-based data visualization library.
- Reflectly — AI Diary & Mood Tracker.
- Zenie AI / Aksha AI — emotion-aware journaling references.

11 — WEBSITE LINK

<https://ranjankashish423-eng.github.io/musemind/>

THANK YOU

MuseMind — AI Personal Journal

Questions & Discussion